

- Least freight charge
- Most ordered items
- Average sales per month/per year

Measures of central tendency

Mean: The mean is the number you obtain when you sum up a given set of numbers and then divide this sum by the total number of samples in the set. The mean is very sensitive to outliers. It can be drastically affected by values that are much higher or lower compared to the rest of the data set.

```
Select b.ContactName,AVG(Freight)Freight
from ordersasa
Join customers as b on a.CustomerID=b.
CustomerID
Group by a.CustomerID;
```

Contact Name	Freight
Maria Anders	37.59667
Ana Trujillo	24.355
Antonio Moreno	38.36
Thomas Hardy	36.30385
Christina Berglund	86.64
Hanna Moos	24.03714
FrdriqueCiteaux	56.69636
Martn Sommer	63.72333
Laurence Lebihan	79.87471

Median: The median is the number separating the higher half of an ordered data set from the lower half. Sometimes the median is a better choice of a mid-point because each data point is weighted equally with respect to the median value. Since MySQL doesn't offer a built-in function to calculate the median value of a column, it is necessary to build a query of our own.

Let us assume that we want to retrieve the median value of the column quantity from the table *orderitem* as shown below:

ID	Order_ID	Prod-uct_ID	Quantity
1	1	1	3
2	2	5	8
3	3	4	9
4	4	6	5
5	5	4	4
6	6	2	9

Let's look into the algorithm we're going to use to build the query.

- Retrieve a sorted list of the values from column *Quantity* and attach an index to each row.
- In case of an odd number of values in the column, find the value of the item in the middle of the sorted list.
- In case of an even number of values in the column, find the values of the two items in the middle of the sorted list.
- Calculate the average of the value(s) retrieved in these steps.
- Return the average value as the median value.

```
1. SET @rowindex:=-1;
2. SELECT
3. Order_ID,rowindex,AVG(Quantity)as
Median
4. FROM
5. (SELECTOrder_ID,@rowindex:=@
rowindex+1 AS rowindex,
6. orderitem.Quantity AS Quantity
7. FROM orderitem
8. ORDER BYorderitem.Quantity) AS g
9. WHERE
10. g.rowindexIN(FLOOR(@
rowindex/2),CEIL(@rowindex/2))
```

Order_ID	rowindex	Median
4	2	7.5

Since there are even numbers of items, the median is an average of

two middle values. The query structure is a little complex. To get a better understanding of the above query, we can explore the sub-query (lines 5-8) of the *FROM* clause of the above query.

```
1. SET@rowindex:=-1;
2. SELECTOrder_ID,@rowindex:=@
rowindex+1ASrowindex,
3. orderitem.QuantityASQuantity
4. FROMorderitem
5. ORDERBYorderitem.Quantity
```

Order_ID	rowindex	Quantity
1	0	3
5	1	4
4	2	7
2	3	8
3	4	9
6	5	9

The output table of the sub-query shows that the median value will be between 7 and 8.

Mode: The mode is the value that appears most often in a set of data. To find the mode of the quantity of the table *orderitem* we can write the mode query as follows:

Mode query

```
SELECT
orderitem.quantity,
COUNT(*) as count
FROM orderitem
Group by orderitem.quantity
ORDER BY COUNT(*) DESC
LIMIT1
```

The *GROUP BY* clause provides a count of the items in each group and the *ORDER BY* clause provides, in descending order, the maximum (mode) count value at the top. The *LIMIT* clause